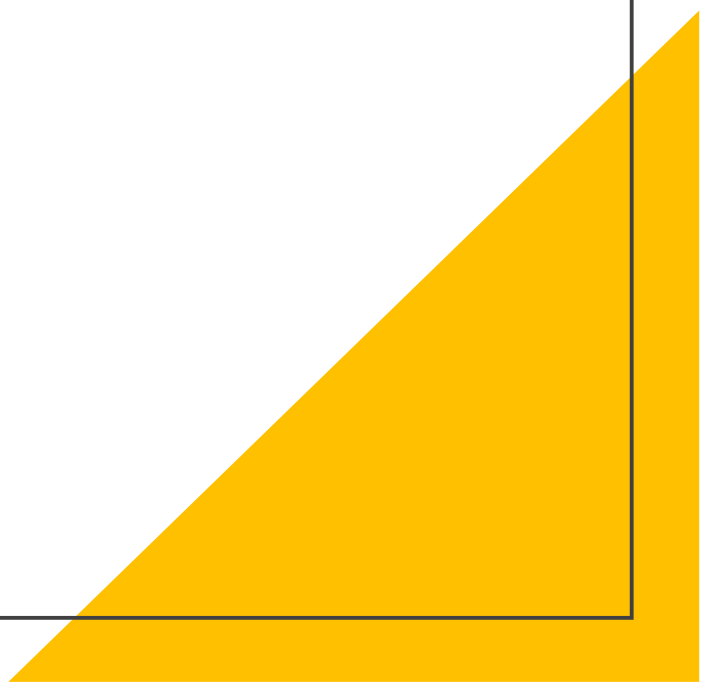


Getting started in Research and Analytics

Session 3: Power analysis (t-test, chi-square)





Power Analysis

In this session,
we'll cover
power analysis.

We'll review
power for t-tests
& chi-square tests



Why should we learn this?

To help plan studies
and answer
challenging questions
by reviewers.

We can answer, “How
big of a sample do you
need?” or “Does this
study have power?”



What is a power analysis?

- A power analysis is a way to calculate the sample size needed to achieve a desired probability that allows us to reject a null hypothesis, if the alternative hypothesis is true. We use the effect size, power level, alpha level, and statistical test to calculate the result.
- A power analysis is also a way to calculate the level of power we have based on our effect size, sample size, alpha level, and statistical test.
- For example, we conduct an experiment and find that Treatment B is better than Treatment A and has a power level of 0.80. If we conduct the same experiment 100 times, we expect to see significant results in about 80 of the 100 trials.

Key concepts

- Power: Goal may be to have 80% power in the analysis given our sample size, alpha level, and expected effect size.
- Effect size: Standardized differences (e.g., Treatment A – Treatment B)
- Alpha level: This is the probability that a researcher expects in making a Type I error. The standard level is 0.05, hence $p < 0.05$.
- Sample size: Goal may be to have a large enough sample to achieve 80% power in the analysis given our alpha level and expected effect size .
- If we know 3 of the 4 values above, we can figure out the 4th value:

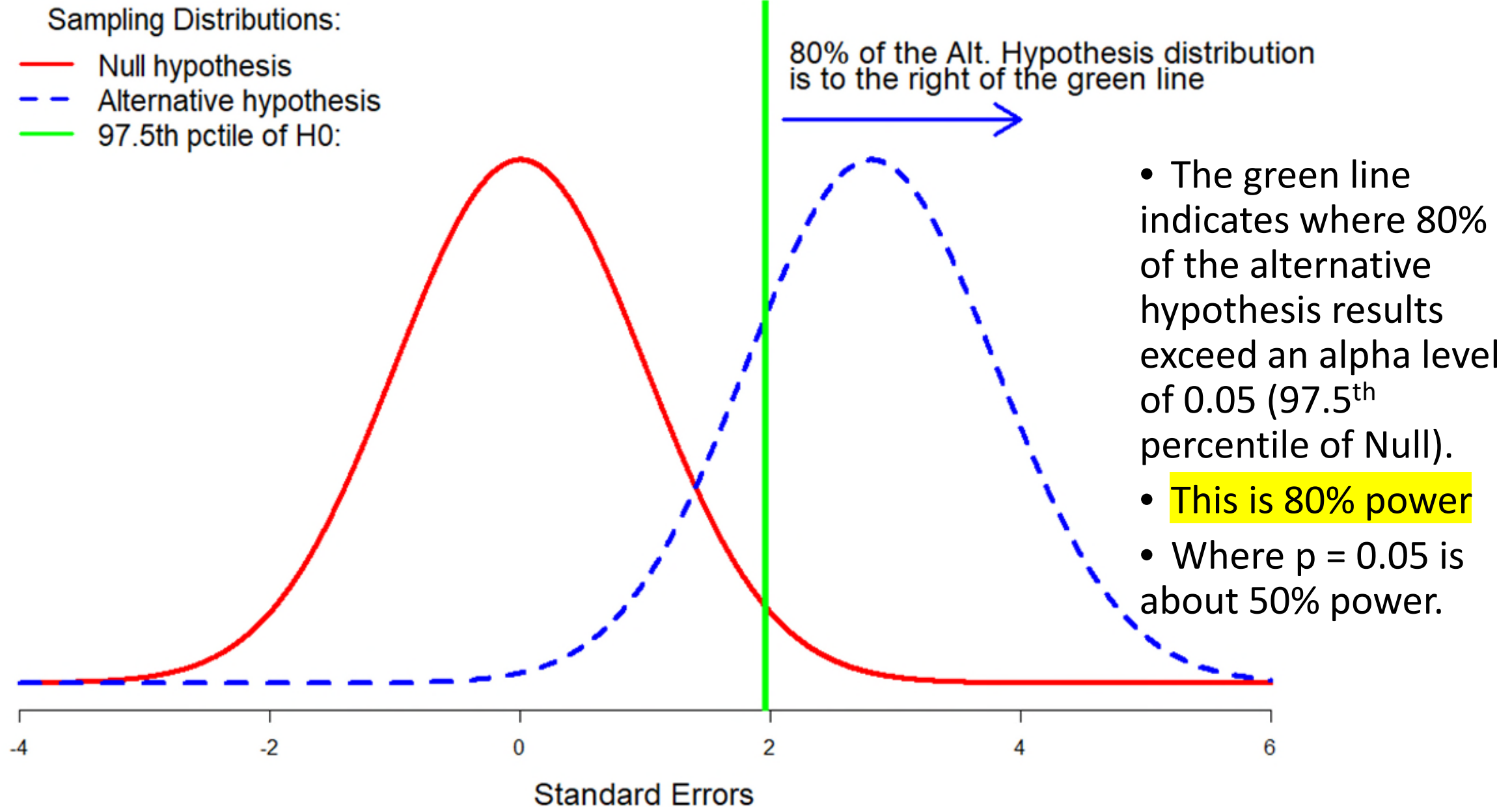
Power <- Effect size, Alpha, Sample Size

Sample Size <- Effect size, Alpha, Power

Types of power analysis tests

- One-sample proportion tests
- Two-sample t-tests
- Correlations
- R^2
- One- or two-tail tests
- Regression coefficients
- Many others inferential statistics

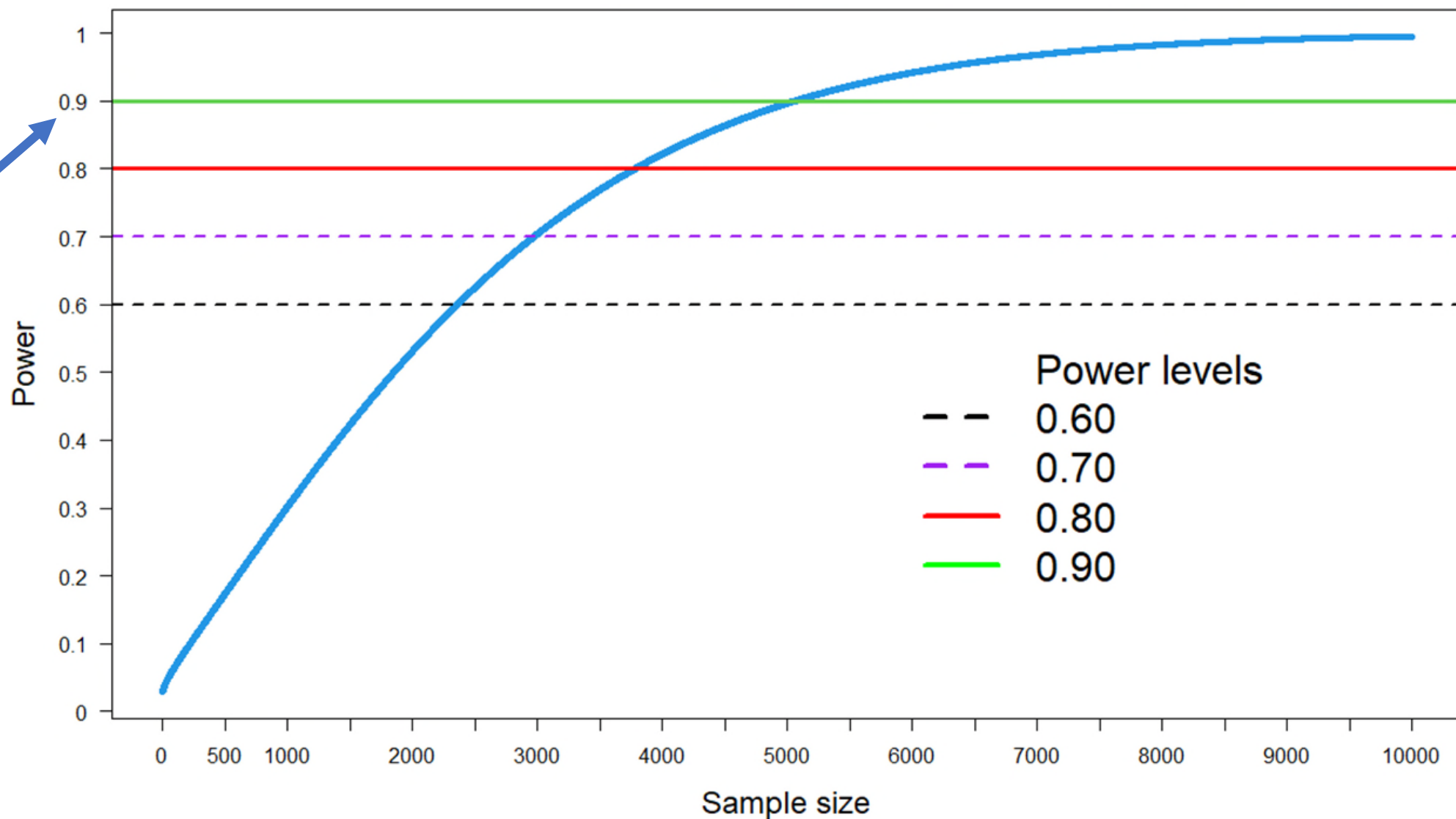
Visualization of 0.80 power



Strategies for increasing power

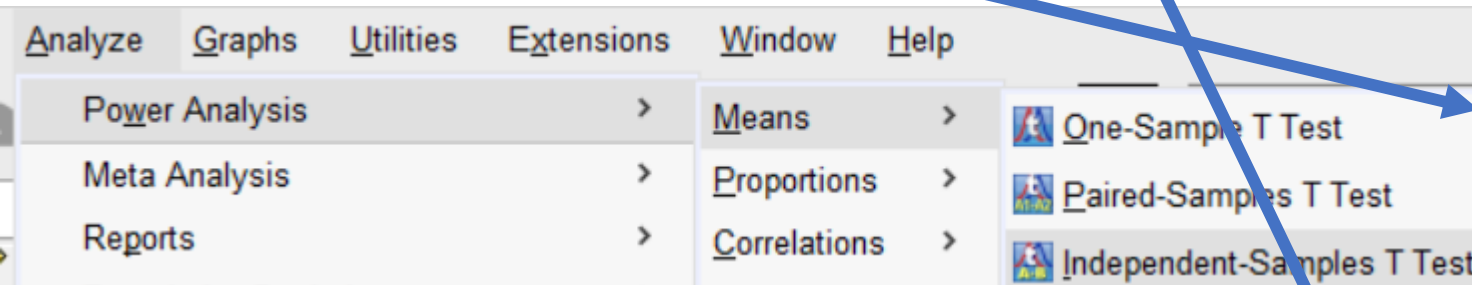
- Sample size
- Add subjects.
- Assign more subject to groups which are cheaper to run.
- Significance level
- Choose a less stringent alpha (or significance) level.
- Effect size
- Increase the size of the hypothesized effect size.
- Employ as few groups as possible.
- Hypothesize main effects rather than interactions.
- Employee measures which are sensitive to change.

Power as a function of sample size



Achieving power beyond 0.90 would require very high sample sizes in this example and all others. This is costly.

SPSS Power Analysis: Sample size Independent Sample t-test 1) Power, 2) Effect, 3) Alpha



Power Analysis: Independent-Sample Means

Test Assumptions

Estimate: Sample size ▾

☒ Single power value: 0.80

☐ Grid power values: Grid

Grid values: None selected

Group size ratio: 1

Specify: Hypothesized Values ▾

☒ Population mean difference: 300

☐ Population mean for group 1: and group 2:

Population standard deviations are

☒ Equal for two groups

Pooled standard deviation: 700

☐ Not equal for two groups

Standard deviation for group 1: and group 2:

Test Direction

☒ Nondirectional (two-sided) analysis

☐ Directional (one-sided) analysis

Significance level: 0.05

Plot Precision

OK Paste Reset Cancel Help

Using values from the data

You can adjust the settings (e.g., unequal variances)

Power Analysis Table						
Test for Mean Difference ^a	Test Assumptions		Actual Power ^b	Power	Std. Dev. ^c	Effect Size
	N1	N2				
Test for Mean Difference ^a	87	87	.803	.8	700	.429

a. Two-sided test.

b. Based on noncentral t-distribution.

c. Group variances are assumed to be equal.

Based on our inputs, we need a sample size of N=174

Same Power Analysis in R and Python

R (see full code for data, uses base library)

```
power.t.test(power=0.80, delta=300,  
sd=700,  
sig.level=0.05,  
type="two.sample",  
alternative="two.sided")
```

Based on our inputs, we need a sample size of N=174 (87 per group)

Python (see full code for libraries/data)

```
import numpy as np  
import statsmodels.stats.power as smp  
# Calculate Cohen's d (Effect Size) Effect/pooled SD  
effect_size = 300 / 700  
# Initialize the power analysis object  
power_analysis = smp.TTestIndPower()  
# Power Analysis  
required_n = power_analysis.solve_power(  
    effect_size=effect_size,    alpha= 0.05,    power=  
0.80,  
    ratio=1.0,    # 1.0 means equal group sizes  
    alternative='two-sided')  
# Required sample size per group, round up  
print(required_n)
```

What about how much power the analysis has? Slightly modify the commands, replace Power with N sample size.

R (same as previous slide)

```
# Formula for harmonic mean
round((2*(74 * 115))/(74 + 115), 0)

# n = 90 = harmonic mean of 74 smokers and 115
non-smokers

power.t.test(n=90, delta=300,
sd=700,
sig.level=0.05,
type="two.sample",
alternative="two.sided")
```

Based on our inputs, our level of power = 0.82

Python (same as previous slide)

```
# Calculate Cohen's d (Effect Size) Effect/pooled SD
effect_size = 300 / 700

# get the harmonic mean for unequal group sizes
round((2 * (74 * 115)) / (74 + 115))

# Initialize the power analysis object
power_analysis = smp.TTestIndPower()

# Power Analysis
required_n = power_analysis.solve_power(
    effect_size=effect_size,    alpha= 0.05,  nobs1=90,
    ratio=1.0,                 # 1.0 means equal group sizes
    alternative='two-sided')

# Required sample size per group, round up
print(required_n)
```

Example 1: Sample size



Question: How big of a sample do we need?



Summary: We want to compare proportions of smoking or non-smoking moms that have babies with low birthweights.



We believe the two groups will have low birth weight rates of 0.25 and 0.41 for non-smokers and smokers.



Using a proportion test: alpha level=0.05, power level=0.80, and the above effect size, we need samples of 135 per each group.

Example 2: Power



Question: How much power do we have?



Summary: We want to compare the proportions of smoking or non-smoking moms that have babies with low birthweights.

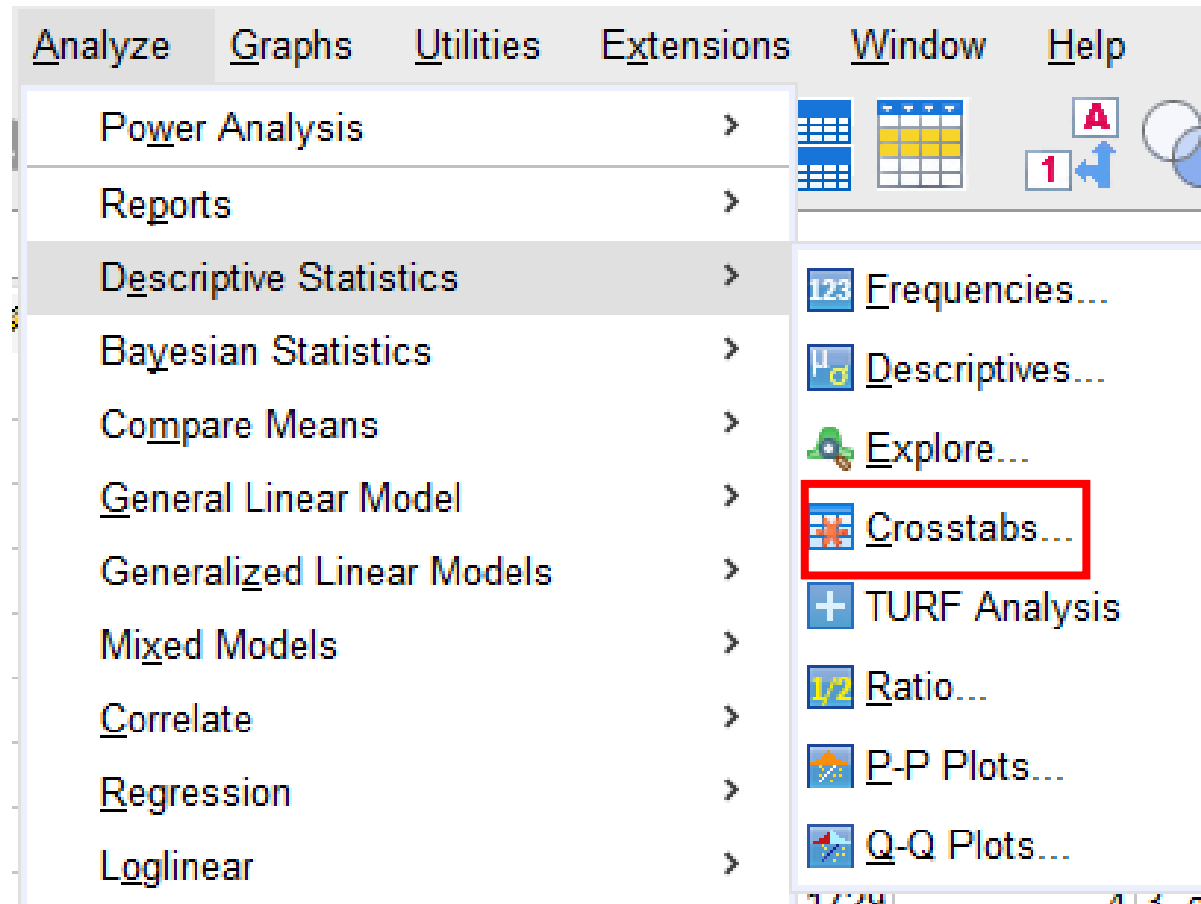


We expect to collect data from 115 non-smokers and 74 smokers.



Using a proportion test with an alpha level=0.05 and an effect size for rates of 0.25 vs. 0.41, we would have a power level=0.64.

1. Calculate crosstabs to get proportions
(steps 1-4 are identical for both examples)



2. Select tests and cell statistics

The image displays three overlapping SPSS Crosstabs dialog boxes, illustrating the configuration of statistical tests and cell display options for a 2x2 contingency table analysis.

Leftmost Crosstabs Dialog:

- Row(s):** low
- Column(s):** smoke
- Layer 1 of 1**
- Display layer variables in table layer:** ☐
- Display clustered bar charts:** ☐
- Suppress tables:** ☐
- Buttons:** OK, Paste, Reset, Cancel, Help

Middle Crosstabs: Statistics Dialog:

- Chi-square:** ☒ (Nominal)
- Contingency coefficient:** ☐ (Nominal)
- Phi and Cramer's V:** ☐ (Nominal)
- Lambda:** ☐ (Nominal)
- Uncertainty coefficient:** ☐ (Nominal)
- Eta:** ☐ (Nominal by Interval)
- Cochran's and Mantel-Haenszel statistics:** ☐
- Test common odds ratio equals:** 1
- Buttons:** Continue, Cancel, Help

Rightmost Crosstabs: Cell Display Dialog:

- Counts:** ☒ Observed, ☒ Expected, ☐ Hide small counts (Less than 5)
- z-test:** ☐ Compare column proportions, ☐ Adjust p-values (Bonferroni method)
- Percentages:** ☒ Row, ☒ Column, ☒ Total
- Residuals:** ☐ Unstandardized, ☐ Standardized, ☐ Adjusted standardized
- Create APA style table:** ☐
- Noninteger Weights:** ☒ Round cell counts, ☐ Round case weights, ☐ Truncate cell counts, ☐ Truncate case weights, ☐ No adjustments
- Buttons:** Continue, Cancel, Help

3. Identify the proportions of (non-)smokers who had babies with low birthweights

low * smoke Crosstabulation

			smoke		Total
			0. No	1. Yes	
low	0. No	Count	86	44	130
		Expected Count	79.1	50.9	130.0
		% within low	66.2%	33.8%	100.0%
		% within smoke	74.8%	59.5%	68.8%
		% of Total	45.5%	23.3%	68.8%
	1. Yes	Count	29	30	59
		Expected Count	35.9	23.1	59.0
		% within low	49.2%	50.8%	100.0%
		% within smoke	25.2%	40.5%	31.2%
		% of Total	15.3%	15.9%	31.2%
Total	Count	115	74	189	
	Expected Count	115.0	74.0	189.0	
	% within low	60.8%	39.2%	100.0%	
	% within smoke	100.0%	100.0%	100.0%	
	% of Total	60.8%	39.2%	100.0%	

Chi-Square Tests

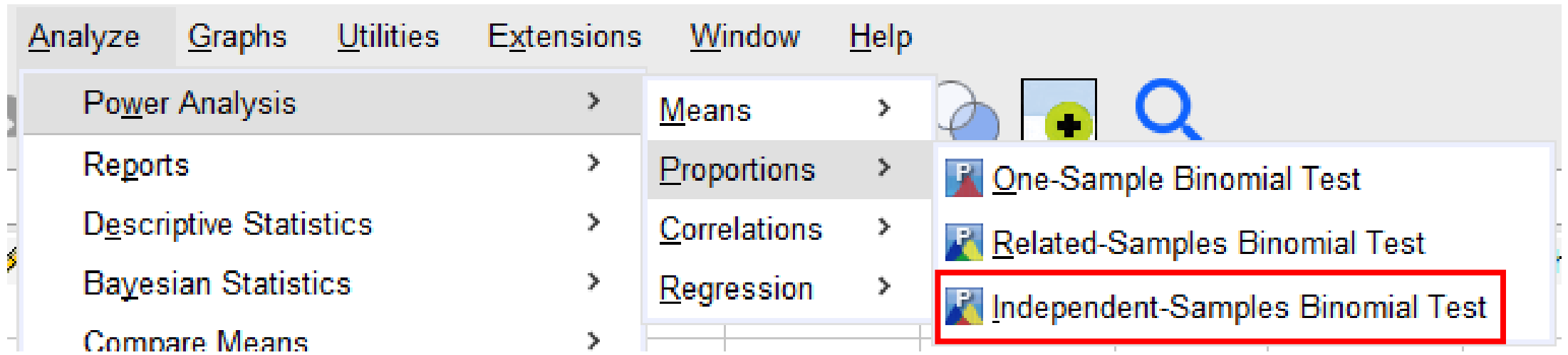
	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	4.924 ^a	1	.026		
Continuity Correction ^b	4.236	1	.040		
Likelihood Ratio	4.867	1	.027		
Fisher's Exact Test				.036	.020
N of Valid Cases	189				

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 23.10.

b. Computed only for a 2x2 table

25% of non-smokers had babies with low birth weights compared to 41% of smokers. All 3 tests above are significant but just barely.

4. Select a power analysis for independent proportions (i.e., all moms are unique cases)



5a. Conduct a power analysis to determine the sample size needed for 0.80 power

Power Analysis: Independent-Sample Proportions

Test Assumptions

☒ Estimate sample size

Power: Group size ratio:

☐ Estimate power

Total number of trials for group 1: and group 2:

Proportion parameters for group 1: and group 2:

Significance level:

Test Method

☒ Chi-squared test

☒ Standard deviation is pooled

☐ Apply continuity correction

☐ I-test

☒ Standard deviation is pooled

☐ Likelihood ratio test

☐ Fisher's exact test

Estimation Method

☒ Normal approximation

☐ Binomial enumeration

☐ Time limit: minutes

Test Direction

☒ Nondirectional (two-sided) analysis

☐ Directional (one-sided) analysis

OK Paste Reset Cancel Help

To achieve 0.80 power with the same results, we would need 135 subjects in each of the 2 groups.

Power Analysis - Independent Proportions

Power Analysis Table								
	N1	N2	Actual Power ^b	Power	Test Assumptions			
					Risk Difference	Risk Ratio	Odds Ratio	Sig.
Test for Proportion Difference ^a	135	135	.802	.8	-.160	.610	.480	.05

a. Two-sided test using large-sample approximation.

b. The estimation of power is based on the Pearson Chi-Square test and the pooled standard deviation.

5b. Conduct a power analysis to determine the power level (caution in post-hoc analyses)

Power Analysis: Independent-Sample Proportions

Test Assumptions

☐ Estimate sample size

Power: Group size ratio:

☒ Estimate power

Total number of trials for group 1: and group 2:

Proportion parameters for group 1: and group 2:

Significance level:

Test Method

☒ Chi-squared test

☒ Standard deviation is pooled

☐ Apply continuity correction

☐ I-test

☒ Standard deviation is pooled

☐ Likelihood ratio test

☐ Fisher's exact test

Estimation Method

☒ Normal approximation

☐ Binomial enumeration

☐ Time limit: minutes

Test Direction

☒ Nondirectional (two-sided) analysis

☐ Directional (one-sided) analysis

OK Paste Reset Cancel Help

Our level of power is 0.64. In other words, we expect to see significant differences repeated in about 64 of 100 future trials.

Power Analysis - Independent Proportions

Power Analysis Table

	Power ^b	N1	N2	Test Assumptions			
				Risk Difference	Risk Ratio	Odds Ratio	Sig.
Test for Proportion Difference ^a	.637	115	74	-.160	.610	.480	.05

a. Two-sided test using large-sample approximation.

b. The estimation of power is based on the Pearson Chi-Square test and the pooled standard deviation.

Summary

Tip: Generally, if you have a p-value of 1) $p < 0.01$, you probably have about 80% power or 2) $p < 0.001$, you probably have about 90% power (**post-hoc power**)

A power analysis can be performed on many types of studies with various types of research questions and research designs.

Alpha, effect size, power, and sample size are directly related. Changing any of those values impacts the other values.

Most researchers aim for a power of .80 to .90. Although, it may be tempting to reach for a power of 1, it will be costly. But oversample if you'll lose subjects.

Caution should be taken in post-hoc power calculations, there is more power when we calculate power before the study rather than after the study.

The effect size should be determined by one of three sources of information: 1) a literature review, 2) results obtained from a pilot study, and 3) ask an expert.

References

- Bausell, R. (2002). *Power analysis for experimental research*. Cambridge, England: Cambridge University Press.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. New York, NY: Taylor & Francis Group.